UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 009 · Particular integral when X is any other form · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·5	Unit 2 deck 009 (“Particular integral when X is any other form”, SESSION 7), slide 5 (rendered .pptx)
T009 00:03:13	Unit 2 AV-summary transcript “Particular Integral when X is any other form” (15.2 min), segment start
T009b 00:01:10	Unit 2 AV-summary transcript “Particular integral when X is any other form_Continued” (10.2 min), segment start
CW U2 P28	Unit 2 class-work problem 28 (pointer only; solved in the class 011 VoP pack)
ESA Dec-23 3c	Past paper: UE22MA141A Dec 2023, question 3.c

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A Formula sheet

1 The general method: linear factors, partial fractions and $\frac{1}{D-m}X$

E1.1 · The standard result for one linear factor ESA ★★

$$\frac{1}{D-m}\,X = e^{mx}\int X\,e^{-mx}\,dx$$

GIVES The PI contribution of one factor $D-m$.

APPLIES Any X; m real or complex.

NOTE Sign: $\frac{1}{D+1}X = e^{-x}\int Xe^{x}dx$ ($m=-1$). No constant of integration.

S009·4 · T009 00:02:11 · T009b 00:03:01 · T009b 00:03:36 · ESA Dec-23 3c

E1.2 · Why: a first-order linear equation ESA ★

$$u = \frac{1}{D-m}X \iff u' - mu = X \iff \left(e^{-mx}u\right)' = Xe^{-mx}$$

GIVES E1.1 by the integrating factor e^{-mx}.

APPLIES Derivation; not on the slides.

NOTE Integrating and multiplying by e^{mx} gives E1.1.

ADDED FOR CLARITY S009·4

E1.3 · Partial fractions of $\frac{1}{f(D)}$ ESA ★★

$$\frac{1}{f(D)} = \frac{A_1}{D-m_1} + \cdots + \frac{A_n}{D-m_n}$$

$$\text{P.I.} = A_1e^{m_1x}\int Xe^{-m_1x}dx + \cdots + A_ne^{m_nx}\int Xe^{-m_nx}dx$$

GIVES The PI as a sum of n single-factor terms.

APPLIES Distinct roots $m_1,\ldots,m_n$.

NOTE Cover-up: $A_k = \frac{1}{f'(m_k)}$; e.g. $\frac{1}{(D+1)(D+2)} = \frac{1}{D+1} - \frac{1}{D+2}$.

S009·4 · S009·10 · S009·12 · S009·14 · T009 00:01:32 · T009 00:01:52

E1.4 · Successive application (no partial fractions) ESA ★★

$$\frac{1}{(D-m_1)(D-m_2)}X = \frac{1}{D-m_1}\left[\frac{1}{D-m_2}X\right] = \frac{1}{D-m_1}\left[e^{m_2x}\int Xe^{-m_2x}dx\right]$$

GIVES The same PI by two single-factor steps, inner factor first.

APPLIES Also with repeated roots ($(D-m)^2$), where partial fractions need an extra term.

NOTE The factors commute; choose the order that makes the integrals easy (W3.1, W3.2).

ADDED FOR CLARITY S009·3 · S009·10 · ESA Dec-23 3c

E1.5 · $m = 0$: integrate ESA ★

$$\frac{1}{D}X = \int X\,dx, \qquad \frac{1}{D^2}X = \iint X\,dx\,dx$$

GIVES The PI when $f(D) = D^k$.

APPLIES $y'' = X$, $y''' = X$, ...

NOTE $\iint \ln x\,dx\,dx = \frac{x^2\ln x}{2} - \frac{3x^2}{4}$ (slide 009·9).

S009·8 · S009·9 · T009 00:12:09

E1.6 · Integrals of $\ln x$ ESA ★

$$\int \ln x\,dx = x\ln x - x, \qquad \int x\ln x\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4}$$

GIVES The two integrations of slide 009·9.

APPLIES By parts with $u = \ln x$.

NOTE Write $\ln x = \ln x \cdot 1$ for the first one.

S009·9 · T009 00:12:37

2 Complex linear factors: $(D^2 + a^2)y = \tan ax$

E2.1 · Complex factors of $D^2 + a^2$ ESA ★★

$$\frac{1}{D^2+a^2} = \frac{1}{(D+ia)(D-ia)} = \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right]$$

GIVES The partial fractions for complex AE roots $\pm ia$.

APPLIES CF $c_1\cos ax + c_2\sin ax$.

NOTE $A = \frac{1}{2ia}$, $B = -\frac{1}{2ia}$ (T009 00:05:02).

S009·5 · S009·6 · T009 00:04:31

E2.2 · Euler's formula

ESA ★

$$e^{\pm iax} = \cos ax \pm i \sin ax$$

GIVES $e^{-iax}\tan ax = \sin ax - i\frac{\sin^2 ax}{\cos ax}$, and $e^{iax}(\cos ax - i\sin ax) = 1$.

APPLIES Inside $\int \tan ax\, e^{\mp iax} dx$.

NOTE $\frac{\sin^2 ax}{\cos ax} = \sec ax - \cos ax$.

S009·6 · T009 00:05:33 · T009 00:07:03

E2.3 · Integrals used

ESA ★

$$\int \sin ax\, dx = -\frac{\cos ax}{a}, \qquad \int \sec ax\, dx = \frac{1}{a}\log(\sec ax + \tan ax)$$

GIVES The two pieces of $\int \tan ax\, e^{-iax} dx$.

APPLIES Slide 009·6.

NOTE Also $\int \cos ax\, dx = \frac{\sin ax}{a}$.

S009·6 · T009 00:07:38

E2.4 · One factor on $\tan ax$

ESA ★★

$$\frac{1}{D \mp ia}\tan ax = -\frac{1}{a}\left[1 \pm i\, e^{\pm iax}\log(\sec ax + \tan ax)\right]$$

GIVES The two partial results (upper signs: slide 009·6; lower signs: 009·7 by $i \to -i$).

APPLIES $(D^2 + a^2)y = \tan ax$.

NOTE Changing i to $-i$ gives the second from the first.

S009·6 · S009·7 · T009 00:08:41 · T009 00:08:58

E2.5 · Result: PI of $\tan ax$

ESA ★★

$$\frac{1}{D^2 + a^2}\tan ax = -\frac{1}{a^2}\cos ax\, \log(\sec ax + \tan ax)$$

GIVES The PI; with the CF, $y = c_1 \cos ax + c_2 \sin ax - \frac{1}{a^2}\cos ax \log(\sec ax + \tan ax)$.

APPLIES Any $a \neq 0$.

NOTE $a = 2$ gives CW P28's answer (asked there by variation of parameters).

S009·7 · T009 00:11:09 · CW U2 P28

3 Right-hand sides in e^x: the substitution $t = e^x$

E3.1 · The substitution $t = e^x$ ESA ★★

$$e^{-mx}\int e^{kx}F(e^x)\,dx = e^{-mx}\int t^{k-1}F(t)\,dt, \qquad t = e^x,\ dt = e^x dx$$

GIVES Integrals of powers of e^x times a function of e^x.

APPLIES $X = e^{e^x}$, $\sin(e^x)$, $e^{2x}\sin(e^x)$, ...

NOTE Write $e^{kx}dx = e^{(k-1)x}\cdot e^x dx = t^{k-1}dt$; substitute back $t = e^x$ at the end.

S009·11 · S009·12 · S009·13 · S009·15 · T009 00:14:05 · T009b 00:03:57 · ESA Dec-23 3c

E3.2 · Integrals in t ESA ★★

$$\int te^t\,dt = (t-1)e^t, \qquad \int t\sin t\,dt = -t\cos t + \sin t$$

$$\int t^3\sin t\,dt = -t^3\cos t + 3t^2\sin t + 6t\cos t - 6\sin t$$

GIVES The by-parts results used on slides 009·11, 009·13, 009·16.

APPLIES After $t = e^x$.

NOTE $\int t^4\sin t\,dt = -t^4\cos t + 4t^3\sin t + 12t^2\cos t - 24t\sin t - 24\cos t$.

S009·11 · S009·13 · S009·16 · T009 00:14:43 · T009b 00:05:16 · T009b 00:08:37

E3.3 · ESA Dec-23 3c: PI of e^{e^x} ESA ★★

$$\frac{1}{D^2+3D+2}\,e^{e^x} = e^{-2x}e^{e^x}$$

GIVES The PI; $y = c_1e^{-x} + c_2e^{-2x} + e^{-2x}e^{e^x}$.

APPLIES $y'' + 3y' + 2y = e^{e^x}$.

NOTE Same as slide 009·10–11; checked by substitution (sympy).

S009·10 · S009·11 · ESA Dec-23 3c

B Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $\frac{1}{D}$ = integration without a constant	E1.1, E1.5
$f(D)$	Operator polynomial	Left-hand operator of $f(D)y = X$	E1.3
X	Right-hand side	A function of x (capital); here not of a standard form	E1.1
m_k	AE root	Root of $f(m) = 0$; $f(D) = \prod(D - m_k)$	E1.1, E1.3
A_k	Partial-fraction coefficient	$\frac{1}{f(D)} = \sum \frac{A_k}{D - m_k}$; $A_k = \frac{1}{f'(m_k)}$ (not an arbitrary constant)	E1.3
a	Constant	In $D^2 + a^2$, $\tan ax$: $m = \pm ia$; in the lecture's $\frac{1}{D-a} f(x)$ it is the root (clash)	E2.1, E2.5
i	Imaginary unit	$i^2 = -1$; $e^{\pm iax} = \cos ax \pm i \sin ax$	E2.2, E2.4
t	Substitution variable	$t = e^x$, $dt = e^x dx$ (not time)	E3.1, E3.2
$\ln x$, $\log x$	Natural logarithm	Both mean $\log_e x$ on these slides	E1.6, E2.3
c_1, c_2	Arbitrary constants	In the CF only	E2.5, E3.3

Remark. Capital X is the right-hand side; small x the variable. $m_1, \ldots, m_n$ are the roots of the auxiliary equation, so $f(D) = (D - m_1) \cdots (D - m_n)$; the transcripts also write the formula with a ($\frac{1}{D-a} f(x) = e^{ax} \int e^{-ax} f(x)\, dx$), and in Topic 2 a is the constant of $D^2 + a^2$, so $m = \pm ia$. $A_1, \ldots, A_n$ are partial-fraction coefficients, not arbitrary constants (c_1, c_2). $t = e^x$ is a substitution variable, not time. $\ln x$ and $\log x$ both mean the natural logarithm (the slides use both).